where z_t is the human's current state, z_{t+1} is the future state resulting from the action, and w denotes the human's policy. Section IV-E describes how we model the human states z^H.

In this formulation, human empowerment directly quantifies how much a human is empowered by taking specific actions, capturing the influence on their future states. This metric is particularly relevant for evaluating social norms, as higher empowerment suggests that the human has more control over their future, aligning better with desirable social interactions. Consequently, optimizing robot policies to maximize human empowerment ensures that the human retains significant influence over their trajectory, facilitating behaviors that enhance the human's agency without assuming any specific goals.

C. Estimating Human Empowerment

Human empowerment, as defined in (2), is expressed through the mutual information between a human's actions and their resulting future states, which is given by:

$$I(a_t; z^H_{t+1}|z^H_t) = \int\int p(z^H_{t+1}, a_t|z^H_t) \times \log\left(\frac{p(z^H_{t+1}, a_t|z^H_t)}{p(z^H_{t+1}|z^H_t)p(a_t|z^H_t)}\right) dz^H_{t+1} da, \tag{3}$$

where $p(z^H_{t+1}, a_t|z^H_t)$ denotes the joint probability of future state and actions, while $p(z^H_{t+1}|z^H_t)$ and $p(a_t|z^H_t)$ are marginal distributions over future states and actions, respectively.

Given the computational intractability of directly calculating this mutual information, we adopt a variational approximation to estimate it. The mutual information can be reformulated using KL divergence between the joint distribution $p(z^H_{t+1}, a_t|z^H_t)$ and the product of the marginal distributions:

$$I(a_t; z^H_{t+1}|z^H_t) = D_{KL}(p(z^H_{t+1}, a_t|z^H_t) \parallel p(z^H_{t+1}|z^H_t)\omega(a_t|z^H_t)), \tag{4}$$

where $\omega(a_t|z^H_t)$ represents the human's policy and $p(z^H_{t+1}|z^H_t)$ models the future state distribution given the current state.

To make this estimation tractable, we employ a variational approximation [25] and introduce neural networks, each with its own parameterization. The pipeline is shown in Fig. 2.

- **Source Policy Network** (ω_{θ_ω}): This network models the human's policy $\omega_{\theta_\omega}(a_t|z^H_t)$, providing the distribution over actions based on the human's current state.
- **Transition Network** (p_{θ_T}): This network models the environment's state transition dynamics $p_{\theta_T}(z^H_{t+1}|z^H_t, a_t)$. It estimates the future states z^H_{t+1} conditioned on the current state z^H_t and action a. It is critical for capturing the influence of a human's actions on their next state.
- **Planning Network** (q_{θ_q}): This network provides a variational approximation $q_{\theta_q}(a_t|z^H_{t+1})$, modeling the posterior distribution over actions given the future state z^H_{t+1}.

Using these networks, we estimate the mutual information by minimizing the KL divergence between the true state transition distribution $p_{\theta_T}(z^H_{t+1}|z^H_t, a_t)$ and the variational approximation $q_{\theta_q}(a_t|z^H_{t+1})$. This leads to the following lower bound for the mutual information:

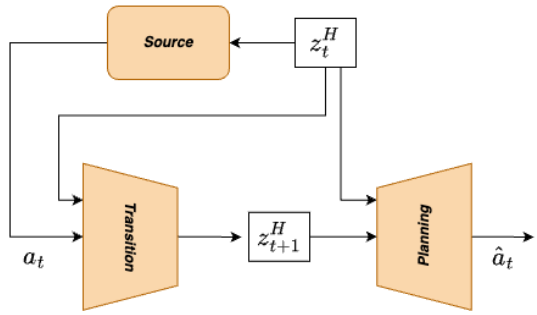

Fig. 2: Network modules of Source, Transition, and Planning used to compute Human Empowerment

$$\hat{I}(a_t; z^H_{t+1}|z^H_t) = \int\int p_{\theta_T}(z^H_{t+1}, a_t|z^H_t) \times \log\left(\frac{q_{\theta_q}(a_t|z^H_{t+1})}{p(a_t|z^H_t)}\right) dz^H_{t+1} da. \tag{5}$$

In this formulation, $q_{\theta_q}(a_t|z^H_{t+1})$ is computed by the Planning Network, while the Transition Network models $p_{\theta_T}(z^H_{t+1}|z^H_t, a_t)$, and the Source Policy Network models $\omega_{\theta_\omega}(a_t|z^H_t)$. To compute this efficiently, we minimize the KL divergence between $p(a_t|z^H_t)$ and $q_{\theta_q}(a_t|z^H_{t+1})$, leading to the following optimization objective:

$$\hat{I}(a_t; z^H_{t+1}|z^H_t) \approx \mathbb{E}_{p_{\theta_T}(z^H_{t+1}, a_t|z^H_t)}\left[\log q_{\theta_q}(a_t|z^H_{t+1}) - \log \omega_{\theta_\omega}(a_t|z^H_t)\right]. \tag{6}$$

By optimizing this objective with respect to the network parameters θ_ω, θ_T, and θ_q, we obtain an efficient estimate of human empowerment, which captures the degree to which a human's actions influence their future states over the course of a trajectory. We use Monte Carlo sampling to handle the high-dimensional integrals involved in this estimation.

D. Training

The training process involves updating the Source Policy Network, Transition Network, and Planning Network through gradient-based optimization. Each network is designed to minimize its own loss function, with the goal of accurately estimating the mutual information between actions and future states, which is then used to compute human empowerment.

The Source Policy Network, parameterized by θ_ω, models the action distribution $\omega_{\theta_\omega}(a_t|z^H_t)$, given the current state z^H_t. The loss function for this network is defined as the negative log-likelihood of the observed actions, formulated as

$$\mathcal{L}_{\text{policy}}(\theta_\omega) = -\mathbb{E}_{(z^H_t, a_t)\sim\mathcal{D}}[\log \omega_{\theta_\omega}(a_t|z^H_t)]. \tag{7}$$

The Transition Network, parameterized by θ_T, is responsible for predicting the next state z^H_{t+1} given the current state z^H_t and action a. Unlike probabilistic models that predict a distribution over possible future states, the Transition Network produces

a point estimate of the next state. Therefore, the training objective for this network is to minimize the **mean square error (MSE)** between the predicted next state and the actual next state. The loss function is expressed as

$$\mathcal{L}_{\text{transition}}(\theta_T) = \mathbb{E}_{(z_t^H, a_t, z_{t+1}^H)\sim\mathcal{D}}\left[\|z_{t+1}^H - \hat{z}_{t+1}^H\|^2\right], \quad (8)$$

where $\hat{z}_{t+1}^H$ represents the predicted next state. This ensures that the network captures the transition dynamics accurately and efficiently minimizes the error between the predicted and true states.

The Planning Network, parameterized by θ_q, approximates the posterior distribution $q_{\theta_q}(a_t|z_{t+1}^H)$, which models the action distribution given the future state. Since the true action distribution $p(a_t|z_t^H)$ is not available, we maximize a variational lower bound (ELBO) [26] on mutual information, which serves as a tractable approximation. The loss function for the Planning Network includes an entropy regularization term, preventing the predicted action distribution from becoming overly deterministic. The loss function is formulated as

$$\mathcal{L}_{\text{planning}}(\theta_q) = -\mathbb{E}_{q_{\theta_q}(a_t|z_{t+1}^H)}[\log \omega_{\theta_\omega}(a_t|z_t^H)] + \lambda\mathbb{H}(q_{\theta_q}(a_t|z_{t+1}^H)), \quad (9)$$

where λ is a regularization coefficient, and $\mathbb{H}$ represents the entropy of the action distribution. By maximizing the variational lower bound, the Planning Network ensures that its predictions maintain sufficient diversity and align with the current action distribution predicted by the Source Network.

Training is performed using mini-batch gradient descent. At each step, we sample mini-batches of state-action-next state triples (z_t^H, a_t, z_{t+1}^H) from the dataset $\mathcal{D}$. The forward pass through the networks computes the respective losses for the Source Policy, Planning, and Transition Networks. Then gradients are computed through backpropagation, and the parameters $\theta_\omega, \theta_T, \theta_q$ are updated iteratively. The training process continues until the total loss converges, indicating that the networks have successfully learned to model the human decision-making process, transition dynamics, and action distribution in future states.

Mutual information between actions and future states is estimated by combining the outputs of the Source Policy and Planning Networks, while the Transition Network contributes to predicting the future state based on the current state and action. The mutual information estimate is given by (6). We summarize the core pipeline in Fig. 2 and Algorithm 1.

E. Occupancy Maps

To model human motion and interactions in dynamic environments, we utilize an *occupancy map* to represent the spatial distribution and movement of humans. The occupancy map is a grid-based structure where each cell provides information about the presence and velocity of a neighboring agent. This representation is essential for capturing both static and dynamic elements of the environment, enabling effective navigation and interaction planning.

The occupancy map is formalized as a 3D tensor with dimensions $c \times r \times 3$, where c and r denote the height and width of the grid. Each cell e_j in the grid contains a tuple $e_j = [\text{occupancy}, v_x, v_y]$:

- *occupancy*: A binary value indicating whether the cell is occupied by a human (1 for occupied, 0 for unoccupied).
- v_x, v_y: The velocity components of the human in the x- and y-directions, respectively.

Each occupancy map is ego-centric, centered around a specific human, and captures the local environment around them. The spatial grid provides an immediate snapshot of the human's surroundings, with each cell representing a discrete location in the vicinity of the human. The inclusion of velocity components within each cell allows the system to model not only the spatial arrangement of neighboring humans but also their movement patterns, which is crucial for planning future actions in dynamic environments.

We denote the occupancy map around human k as $g_k(i,j) = \begin{bmatrix}\text{occupancy}_{ij} & v_{x,ij} & v_{y,ij}\end{bmatrix}$: Here, i and j represent the row and column indices of the grid, defining the spatial position relative to the human. The complete human state at time t, denoted z_t, is a concatenation of the individual occupancy maps of all neighboring humans as $z_t^H = [g_1, g_2, \ldots, g_k]$

where each g_k provides a local view of the environment surrounding a particular human. This concatenated structure enables the model to reason about the interactions and potential collisions between multiple humans in the vicinity, facilitating accurate decision-making for navigation tasks.

Algorithm 1 Human Empowerment

Initialize Networks : Source Network (θ_ω), Transition Network (θ_T), Planning Network (θ_q)
Initialize dataset $\mathcal{D}$
while not converged **do**
  Sample a batch $< s_t, s_{t+1} >$ from $\mathcal{D}$
  Compute occupancy map z^H from state s
  Sample action from source policy $a_t \sim \omega_{\theta_\omega}(.|z_t^H)$
  Find future map z_{t+1}^H from transition network $z_{t+1}^H = p_{\theta_T}(.|z_t^H, a_t)$
  Compute losses: $\mathcal{L}_{\text{policy}}$, $\mathcal{L}_{\text{transition}}$, and $\mathcal{L}_{\text{planning}}$
  Update gradients:
  $\Delta\theta_\omega \propto \nabla_{\theta_\omega} \log \omega_{\theta_\omega}(a_t|z_t^H)$
  $\Delta\theta_T \propto \nabla_{\theta_T} \mathcal{L}_{\text{transition}}$
  $\Delta\theta_q \propto \nabla_{\theta_q} \log q_{\theta_q}(a_t|z_{t+1}^H)$
end while
Compute empowerment

$$E(s) = \left[\log q_{\theta_q}(a_t|z_{t+1}^H) - \log \omega_{\theta_\omega}(a_t|z_t^H)\right].$$

V. Experiments and Results

We use a modified version of the CrowdNav simulation environment [27] to evaluate the performance of different social navigation methods on the metric of human empowerment. The environment allows us to control variables difficult to standardize in human studies, such as crowd density and movement patterns. Widely used to validate crowd navigation

policies, CrowdNav provides a safe, cost-effective, and replicable platform to refine strategies in diverse, simulated scenarios. It enables testing of methods based on metrics like collision avoidance, trajectory smoothness, and advanced measures such as discomfort and empowerment. This controlled setting facilitates large-scale experimentation, offering insights into robot behavior in human-centric spaces while saving resources and mitigating risks. Promising strategies can be prioritized for subsequent human testing.

We examine 4 robot navigation methods in particular - ORCA based on Reciprocal Avoidance [28] with no safety space, Linear which assumes that the robot has a constant velocity from its origin to its destination, SARL - a value-based Reinforcement Learning (RL) method based on the attention mechanism [27], and SAC - a Soft Actor-Critic RL algorithm. We select these methods to capture a range of navigation strategies, from basic trajectory optimization to learning-based methods with varying levels of social-awareness, for initial insights into how empowerment varies. However, the pipeline can adapt to any policy with available trajectory data.

We utilize the circle-crossing scenario, where both robots and simulated humans start from a circular arrangement and attempt to cross the circle to reach a diametrically opposite point. The simulated humans also use ORCA for their navigation. This scenario creates a challenging environment as humans and robots converge toward the center, requiring collision avoidance and socially compliant navigation strategies.

We seek primarily to answer the following questions.

- Does the empowerment metric validate our intuitions about how humans perceive their autonomy evolving in dynamic social environments?
- Is the metric capable of capturing differences across different navigation policies?
- How does the empowerment metric compare to other social conformity metrics, such as discomfort?

We believe that from positive responses to these questions, we can demonstrate the utility of empowerment as a metric for evaluating human-robot interactions in social navigation tasks.

A. Temporal Dynamics of Human Empowerment

Fig. 3 illustrates the evolution of human empowerment over time for three navigation methods: ORCA, SAC, and SARL. Initially, empowerment is high, indicating that the human has considerable control over their actions. However, as humans approach closer to each other near the center of the circular environment, empowerment decreases across all methods. This reduction is expected, as proximity to other agents restricts a human's ability to move freely, diminishing their control and influence over future states. After a few more time steps, empowerment values recover as the humans regain autonomy. This pattern confirms that empowerment effectively captures how an agent's autonomy evolves in response to dynamic changes in social environments.

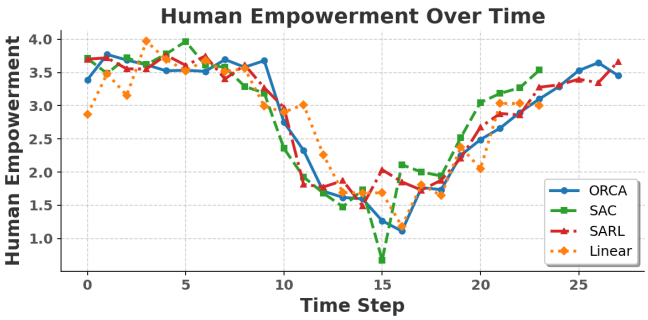

Fig. 3: Empowerment for Different Policies vs. Time

B. Impact of Crowd Density on Human Empowerment

Fig. 5 shows how human empowerment decreases as the number of humans in the environment increases, across the different navigation methods mentioned previously. Initially, with fewer humans, empowerment is relatively high, reflecting the agents' greater autonomy and ability to influence their environment. However, as crowd density increases, empowerment steadily declines across all methods, indicating a reduced capacity to act freely in denser crowds. This aligns with expectations, as higher crowd densities impose more constraints on human movement, requiring more cautious navigation to avoid collisions and respect personal space. The overall downward trend confirms that empowerment effectively captures an agent's reduced freedom to navigate in dense environments, with minor differences between methods showing slight variations in performance.

C. Human Empowerment Across Policies

We provide violin plots displaying the distribution of mean empowerment values across all trials (500 seeds) for each policy, as shown in Fig. 6. The figure illustrates the central tendency and spread of empowerment values, allowing for direct comparison between policies. In addition, to provide a clearer understanding of performance in terms of success, we present the success rates for each policy in Table I. Notably, we include all trials (both successful and unsuccessful) in the empowerment analysis. Unsuccessful trials are defined as those where a collision occurred, or the agent failed to reach the goal within the specified time. If a collision occurs, the episode ends prematurely. Since we report the mean of average empowerment values over the entire episode and across all humans, early terminations due to collisions may not significantly impact the overall mean empowerment. This explains why scenarios with lower success rates can still exhibit higher empowerment values. Note that these two metrics are independently useful in assessing policy performance.

TABLE I: Success rate for different policies

Policy	ORCA	SAC	SARL	Linear
Success	0.456	0.99	0.44	0.05